SOLSTICE Task2B

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§ MATHEMATICAL Setup: QR Code Rectifier

The Ising Model Setup:

- Prior [Coupling] $\Rightarrow$ Pixels in QR tend to be same colour as Neighbours
- Likelihood [Fidelity] $\Rightarrow$ True Pixel $\simeq$ Noisy Observation (if Noise isn't overwhelming)

1. PRIOR

$$P(Z) = \frac{1}{Z_{\text{Prior}}} e^{\beta \sum_{(i,j)\in E} Z_i Z_j}, \quad Z_{\text{Prior}} = \sum_{\text{All } Z} e^{\beta \sum Z_i Z_j}$$

Where:

- $Z_i = \pm 1 \quad$ (Pixel Value) $\rightarrow$ current Guess
- $E \rightarrow$ all Adjacent pairs
- $\beta \rightarrow$ Hyper parameter
- $Z_{\text{Prior}} \Rightarrow$ Sum of All possible Theoretical Configuration of the Entire Grid Z
- $Z = [Z_1, Z_2, \ldots, Z_{N^2}]$
- $Z_1 Z_2$: Same Pixel is +ve

Clearly if:

- $\beta > 0 \Rightarrow$ Encourages Same Valued Neighbors
- $\beta < 0 \Rightarrow$ Encourages Alternating Patterns

Likelihood: Trusting Noisy image.

[If my current guess for True Pixel (z) is correct, how likely is it that the Noise would produce the dirty Pixel (x). I am looking for]

$$P(X|Z) = \prod_i P(x_i|z_i) = \frac{1}{Z_{\text{like}}} e^{\eta \sum_i X_i Z_i}$$

Where:

- $Z_i \Rightarrow$ True Clear image
- $X_i \Rightarrow$ Noisy observed image
- $\eta \uparrow \Rightarrow$ Trust the Noisy Data Lot
- $\eta \downarrow \Rightarrow$ Trust the data less

Q2 Bayes:

$$P(Z|X) \propto P(X|Z) \cdot P(Z)$$
$$\Rightarrow P(Z|X) = \frac{1}{Z_{\text{Post}}} \cdot e^{\left[\beta \sum_{(i,j)\in E} Z_i Z_j + \eta \sum_i x_i z_i\right]}$$

Q3 β, η are Hyper Parameters.

$$P(\text{unobserved} \mid \text{observed}) \propto P(\text{observed} \mid \text{unobserved}) \times P(\text{unobserved})$$
$$\Rightarrow P(Z, \beta, \eta \mid X) \propto P(X \mid Z, \beta, \eta) \times P(Z, \beta, \eta)$$

Now,

- $P(Z, \beta, \eta) = P(Z \mid \beta, \eta) \cdot P(\beta, \eta)$
- $P(\beta, \eta) = P(\beta) \cdot P(\eta)$ [as β, η are independent]
- $P(Z \mid \beta, \eta) = P(Z \mid \beta)$ [because True model doesn't care about Noise]
- $P(X \mid Z, \beta, \eta) = P(X \mid Z, \eta)$ [because β is irrelevant as all that info already there in Z]

Thus,

$$P(Z, \beta, \eta \mid X) \propto \underbrace{P(X \mid Z, \eta)}_{\text{Likelihood}} \times \underbrace{P(Z \mid \beta) \times P(\beta)P(\eta)}_{\text{Prior}}$$

PROBLEM 1 QR code: Implementation

Q: What noise are you adding and why?
We are adding independent bit-flip noise. We use it because it directly maps to the mathematical structure of the variables (binary states of +1 and -1).

Q: How does this affect the reasoning behind your prior and likelihood?
The likelihood equation we used completely relies on the assumption that the noise is Independent and Identically Distributed (i.i.d.). This means the probability of pixel A flipping has zero physical effect on whether the adjacent pixel B flips. Thus independent bit-flip noise completely justifies our choice of Likelihood equation.

Sampling Method: Gibbs Sampling.
Reason: (the pixels) are strictly discrete, binary values (-1 for Black, 1 for White) {rules out HMC which requires continuous surface}. Our chosen prior (Ising) has local dependencies on neighbourhood. Gibbs Sampling is perfectly designed for this exact architecture: it freezes the global grid and iteratively updates individual nodes based strictly on their local Markov blanket (their neighbors) and the corresponding likelihood from the observed noisy image.

2 (a)

Total System Energy (the unnormalized negative log-posterior).
We track the Energy of the grid after every sweep. If the sampler is functioning correctly, it should be moving toward states of higher probability, which mathematically translates to minimizing the system's total energy.

$$E_{total} = -\left(\eta \sum_i X_i Z_i + \beta \sum_{\langle i,j \rangle} Z_i Z_j\right)$$

2 (b)

Cross-validation loss evaluates how well a model predicts a hidden set of ground truth data. However, Bayesian image denoising is an unsupervised problem. Thus we cannot use CV loss.

2 (c)

Limitations: The primary limitation of Gibbs Sampling in this context is "slow mixing" when variables are highly correlated. Because it only updates one pixel at a time, it struggles to make large-scale structural changes if it gets trapped in a local optimum. If the hyperparameter Beta (prior strength) is set too high relative to eta (likelihood confidence), the algorithm will suffer as it ignores the noisy evidence entirely just to satisfy the rule that neighbors must match.

To detect this, we track the ratio of white to black pixels (or the Mean Pixel Value). A valid QR code is roughly 50% white and 50% black (a mean near 0). If this tracked metric rapidly diverges toward 1 (pure white) or -1 (pure black) during sampling, the model has collapsed, and the hyperparameters must be retuned.

2 (d)

The burn-in phase is the initial transient period where the Markov Chain moves from its highly randomized starting state (the noisy image) toward the high-probability equilibrium distribution. The samples generated during this phase are inaccurate and must be discarded before calculating final averages.

We track the Total System Energy across sweeps. Plotting this metric will show a steep, rapid drop as the obvious noise is cleared. The exact sweep where this curve flattens out into a noisy, horizontal plateau marks the end of the burn-in phase.

PROBLEM 2: Bots vs Humans

Mathematical Setup: Ising Prior

1. Prior (Ising Prior)

Assumption: bots follow bots, humans follow humans

$$P(Y) = \frac{1}{Z_{\text{Prior}}} \cdot e^{\beta\left(\sum_{(i,j)\in E} Y_i Y_j\right)}$$

2. Likelihood

$$P(A|Y) = \frac{1}{Z_{\text{likelihood}}} e^{\eta\left(\sum_{i\in V} A_i Y_i\right)}$$

$$\text{Total Energy} = -\left[\eta \sum A_i Y_i + \beta \sum Y_i Y_j\right]$$

$$P(Y|A) = \frac{1}{Z_{\text{Post}}} e^{-\text{Total Energy}}$$

Implementation

The Mathematical setup is very similar to previous problem.
We generated a synthetic network using a Stochastic Block Model (networkx.random_partition_graph). The noise added is an independent bit-flip on the ground-truth nodes.
Sampling method: Gibbs Sampling traversing an Adjacency List.

(a) To ensure the sampler is functioning without cheating by looking at the True labels, we track the Total System Energy.

(b) we cannot calculate a standard Cross-Validation loss as mentioned in problem 1.

(c) Limitation: In the QR grid, every pixel had 4 neighbors. In a social graph, a "Lurker" has 1 neighbor, while an "Influencer" has 10,000. If we use standard Gibbs, the Prior term (beta) for the Influencer becomes massive, completely destroying the Likelihood(eta). A few bot followers could mathematically force a human influencer to be labeled a bot.

To prevent this issue: We can track the Mean Node Value across sweeps. If the Influencer effect cascades and crashes the model, the mean value will rapidly diverge toward +1 (everyone is a bot) or -1 (everyone is human).

§ Code Notebook file

Click Here to Open the Notebook